

xjyutping: Jyutping above traditional Chinese characters

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Abstract

xjyutping typesets Jyutping (粵拼), the Linguistic Society of Hong Kong romanisation of Cantonese, above Chinese characters. Readings are chosen from the surrounding words, so 行 is *hong4* in 銀行 and *haang4* in 行路. Every character sits in a cell of the same width, wide enough for its Jyutping, so the text stays evenly spaced and no Jyutping touches its neighbours or the line above. The package works with Xe_{La}T_EX and xeCJK, and with the ctex classes.

1 Getting started

```
\documentclass{article}
\usepackage{xeCJK}
\setCJKmainfont{Songti TC}
\usepackage{xjyutping}
\begin{document}
\begin{jyutpingscope}
我哋去銀行，行路返屋企。校長喺學校長大。
\end{jyutpingscope}
\end{document}
```

ngo5 dei6 heoi3 ngan4 hong4 haang4 lou6 faan1 uk1 kei2 haau6 zoeng2 hai2 hok6 haau6 zoeng2 daai6
我 哋 去 銀 行， 行 路 返 屋 企。 校 長 喺 學 校 長 大。

If xeCJK is not loaded yet, xjyutping loads it. Any traditional Chinese font will do; the Jyutping uses the document's main Latin font unless you choose another.

2 Commands

2.1 Annotating text

`\begin{jyutpingscope}[\langle options \rangle] ... \end{jyutpingscope}` annotates every Chinese character in the block. The block ends the paragraph, and its line spacing is raised where needed so that the Jyutping of one line never comes close to the line above.

`\xjyutping*[\langle options \rangle]{\langle text \rangle}` does the same for a piece of running text inside an ordinary paragraph.

香港人講\xjyutping*{廣東話}，寫\xjyutping*{繁體字}。

gwong2 dung1 waa2 faan4 tai2 zi6
香 港 人 講 廣 東 話， 寫 繁 體 字。

2.2 Choosing a reading yourself

Cantonese pronunciation shifts with meaning and with colloquial tone change, and no word list is perfect. There are three ways to say what you want.

`\xjyutping[\options]{\characters}{\readings}` gives the characters an explicit reading, here and nowhere else. It works inside a scope and on its own, and it also marks a word boundary.

`\setjyutping{\character}{\reading}` changes the reading a character gets when it is not part of a known word.

`\setjyutping{\word}{\readings}` adds a word, or overrides the reading of a known one. Words always win over single-character defaults.

Readings are Jyutping syllables separated by spaces, one per character. `\setjyutping` is global and takes effect from where it appears, so it can go in the preamble or in the middle of a scope. In the next example 重話 means ‘also said’ (*zung6*, not the default *cung5* ‘heavy’), and 家行 is a name read *gaa1 hang4*:

```
\setjyutping{重話}{zung6 waa6}  
\begin{jyutpingscope}  
佢重話\xjyutping{家行}{gaa1 hang4} 聽日嚟。  
\end{jyutpingscope}
```

keoi5 zung6 waa6 gaa1 hang4 ting1 jat6 lai4
佢 重 話 家 行 聽 日 嚟。

2.3 Formatting inside words

Braces and formatting commands do not split a word, so a character can be highlighted without losing its reading in context:

```
\begin{jyutpingscope}  
佢嚟銀\textbf{行} 返工，係校\textcolor{red}{長} 嘅朋友。  
\end{jyutpingscope}
```

keoi5 hai2 ngan4 hong4 faan1 gung1 hai6 haau6 zoeng2 ge3 pang4 jau5
佢 嚟 銀 行 返 工， 係 校 長 嘅 朋 友。

2.4 Switching off

`\disablejyutping` stops annotation until `\enablejyutping` or the end of the current group or environment. Use it for passages that should stay plain inside a scope.

3 How readings are chosen

1. The text is split into runs of Chinese characters. Punctuation, Latin text and most commands end a run. Spaces and line breaks between two characters do not, and neither do braces or formatting commands (`\textbf`, `\emph`, `\color`, `\textcolor`, size commands ...). The text of a `\footnote` is a run of its own, and the text around the footnote mark carries on.

2. Each run is divided into words from a list of about 100 000 Cantonese words: the division with the fewest words wins, then the one with the fewest single characters, and on a tie the longer final word.
3. A word takes its reading from the list. Your `\setjyutping` words are checked first.
4. A character that is not part of a word takes your `\setjyutping` reading if you gave one, otherwise its default reading. A few characters read differently at the end of a run: 呢 is *nei1* in 呢張床 but the particle *nei1* in 你呢?
5. `\xjyutping{characters}{readings}` always wins, and ends the run.

Traditional characters have several accepted shapes (為 and 爲, 裡 and 裏, 說 and 説, 衛 and 衛, 線 and 綫, 溫 and 温 ...). For word lookup they are folded together, so a word is found whichever shape you type.

3.1 Proofreading

About four thousand characters have more than one common reading. When such a character is not settled by a word or by you, its reading is only a guess. The `multiple` option formats those guesses, and the `debug` option writes every run to the log with the alternatives:

```
\xjyutpingsetup{multiple=\color{red}, debug}
\begin{jyutpingscope}
佢重未嚟。
\end{jyutpingscope}
```

keoi5 cung5 mei6 lai4
佢 重 未 嚟。

The log then contains a line such as

```
xjyutping> 佢 keoi5:s | 重 cung5:m(zung6 cung4) | 未 mei6:s | 嚟 lai4:m(lei4) |
```

Each character is followed by its reading and a type: **w** from the word list, **u** set by you, **m** a guessed polyphone (other readings in brackets), **s** a character with a single common reading. Here 重 means ‘still’, so `\setjyutping{重未}{zung6 mei6}` fixes it (and `\setjyutping{重}{zung6}` would make *zung6* the default for 重 on its own).

4 Options

Options can be given to `\usepackage`, to `\xjyutpingsetup`, or in the optional argument of `jyutpingscope` and `\xjyutping`.

Key	Default	Meaning
<code>ratio</code>	0.45	Jyutping size relative to the text
<code>vsep</code>	1.05em	height of the Jyutping baseline above the character’s
<code>hsep</code>	0.15em plus 0.4em	space between cells: the least gap between two Jyutping, plus stretch for justified lines
<code>width</code>	auto	cell width: <code>auto</code> , <code>natural</code> or a length
<code>font</code>	<code>\normalfont</code>	font of the Jyutping (font commands only)
<code>format</code>		extra formatting, e.g. <code>\color{gray}</code>
<code>multiple</code>		formatting for guessed polyphones
<code>fancy</code>	false	tones as pitch strokes (section 5)
<code>debug</code>	false	log every run

4.1 Cell width

With `width=auto` every character in a block gets the same cell, wide enough for the longest Jyutping in that block. With a length you get a fixed grid across the whole document; a Jyutping longer than the cell widens its own cell rather than overlap. `width=natural` makes each cell only as wide as its own character or Jyutping: tighter, but uneven.

```
\xjyutping*[width=auto]{香港中文大學}
\xjyutping*[width=natural]{香港中文大學}
\xjyutping*[width=2em]{香港中文大學}
```

hoeng1 gong2 zung1 man4 daai6 hok6
香 港 中 文 大 學
hoeng1 gong2 zung1 man4 daai6 hok6
香 港 中 文 大 學
hoeng1 gong2 zung1 man4 daai6 hok6
香 港 中 文 大 學

4.2 Size and font

```
\xjyutping*[ratio=0.6, font=\sffamily, format=\color{gray}]{早晨，食咗飯未？}
```

zou2 san4 sik6 zo2 faan6 mei6
早 晨， 食 咗 飯 未？

5 Fancy tones

With the `fancy` option, `\usepackage[fancy]{xjyutping}`, each tone number is shown the way Visual Jyutping shows it: a small stroke that traces the pitch of the tone, then a smaller tone number, raised for the two high tones and lowered for the others. The strokes follow Visual Jyutping's symbols: tone 1 high, tone 2 rising to high, tone 3 mid, tone 4 falling to low, tone 5 rising from low, tone 6 low.

```
{\Large\xjyutping*[fancy]{詩史試時市事}}
\begin{jyutpingscope}[fancy]
我哋去銀行，行路返屋企。校長喺學校長大。
\end{jyutpingscope}
```

si⁻¹ si^{/2} si⁻³ si^{\4} si^{/5} si⁻⁶
詩 史 試 時 市 事
ngo^{/5} dei⁻⁶ heoi⁻³ ngan^{\4} hong^{\4} haang^{\4} lou⁻⁶ faan⁻¹ uk⁻¹ kei^{/2} haa^{u-6} zoeng^{/2} hai^{/2} hok⁻⁶ haa^{u-6} zoeng^{/2} daai⁻⁶
我 哋 去 銀 行， 行 路 返 屋 企。 校 長 喺 學 校 長 大。

The strokes are drawn, not taken from a font, so they work with any `font` and take the colour of `format` and `multiple`. The spacing follows the new shape: cells are measured with the strokes and small numbers in place (so `width=auto` cells come out a little wider), line spacing makes room for the raised numbers, and if the lowered numbers reach further down than the letters in the chosen font, the Jyutping is lifted by the difference so that it keeps its distance from the character. Like every option, `fancy` can be switched on or off for one scope: `\begin{jyutpingscope}[fancy=false]`. A syllable given without a tone number (`\xjyutping{唔}{m}`) is printed as it is.

6 Limits

- The body of `jyutpingscope` and the argument of `\xjyutping*` are read in full before anything is typeset, like any argument, so `\verb` and verbatim environments cannot go inside (put them outside the scope), and in a `\url` or `\href` inside a scope a `%` must be written `\%`.
- `\input{<file>}` inside a scope reads the file as part of the scope, with the same restrictions as the body; a file that uses `\endinput`, `\verb`, verbatim environments, `\makeatletter` or catcode changes is input normally instead. Such files, text that comes from a macro, an `\included` file or `\maketitle` are annotated character by character, without word context; the debug log marks them `[no context]`. Put `\xjyutping*` inside such a macro, or a scope inside the included file.
- The arguments of `\label`, `\ref`, `\cite`, `\index`, `\url`, the first argument of `\href`, the optional argument of `\hyperref`, `\hyperlink`, `\pdfbookmark`, `\includegraphics`, the `cleveref` commands and environment names are passed through untouched.
- Give other commands their Chinese argument in braces (`\textbf{行}`}, not `\textbf 行`).
- Section titles, captions and footnotes inside a scope are annotated. The table of contents is annotated only if `\tableofcontents` is itself inside a scope (an explicit `\xjyutping` in a title is annotated there too); running heads and PDF bookmarks are always plain.
- In `beamer` a `\frametitle` inside a scope stays plain (it is typeset after the scope has ended): write `\frametitle{\xjyutping*{...}}`.
- The underline and emphasis-mark commands of `xeCJKfntef` do not keep the cell spacing inside a scope; use `\underline` or `ulem`'s `\uline` there.
- A scope can hold well over 100 000 characters, but one paragraph is limited to about 15 000 by `TEX`'s memory (about 13 000 with `fancy`).
- A `\setjyutping` inside a scope is noticed while the text is read, so one inside `\iffalse...\fi` or an unused macro definition still applies to the text after it.
- Characters missing from the data are typeset in their cell without Jyutping.
- Only `XeLATEX` is supported.

7 Data and licences

The readings come from the LSHK *Cantonese Pronunciation List of Characters for Computers* (粵拼表) and from the rime-cantonese dictionaries, both under CC BY 4.0; the variant-shape map comes from OpenCC (Apache-2.0). The `fancy` tone marks follow Visual Jyutping (MIT) by Vincent Tam. `tools/build-data.py`, at the root of the source repository, regenerates `xjyutping-chars.def` and `xjyutping-words.def` from these sources; its `CURATED_DEFAULTS`, `CURATED_FINALS`, `CURATED_VARIANTS` and `CURATED_WORDS` tables hold the hand-checked corrections. The package code is under the L^AT_EX Project Public License 1.3c.